To prevent unauthorized access, Firewalls, **Intrusion Detection Systems** (IDS), Virus and Content Scanners, Patches and Hot fixes are used.

CRACKING

Cracking is where someone edits a program source so that the code can be exploited or modified. A cracker (also called a black hat or dark side hacker) is a malicious or criminal hacker. "Cracking" means trying to get into computer systems in order to steal, corrupt, or illegitimately view data.

A cracker is someone who breaks into someone else's computer system, often on a network, bypassing passwords or licenses in computer programs.

They may send official e-mail requesting some sensitive information. It may look like a legitimate e-mail from bank or other official institution.

17.3 Cyber Security and Threats

Cyber attacks are launched primarily for causing significant damage to a computer system or for stealing important information from an individual or from an organization.Cyber security is a collection of various technologies, processes and measures that reduces the risk of cyber attacks and protects organizations and individuals from computer based threats.

TYPES OF CYBER ATTACKS

Malware is a type of software designed through which the criminals gain illegal access to software and cause damage. Various types of cyber-attacks and their functions are given in**Table 17.2.**

Table 17.2 – Cyber Attacks and Functions

S.No.	Cyber Attack	Function
1.	Virus	A virus is a small piece of computer code that can repeat itself and spreads from one computer to another by attaching itself to another computer file. One of the most common virus is **Trojan**. A Trojan virus is a program that appears to perform one function (for example, virus removal) but actually performs malicious activity when executed.
2.	Worms	Worms are self- repeating and do not require a computer program to attach themselves. Worms continually look for vulnerabilities and report back to the author of the worm when weaknesses are discovered.
3.	Spyware	Spyware can be installed on the computer automatically when the attachments are open, by clicking on links or by downloading infected software.
4.	Ransomware	Ransomware is a type of malicious program that demands payment after launching a cyber-attack on a computer system. This type of malware has become increasingly popular among criminals and costs the organizations millions each year.

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